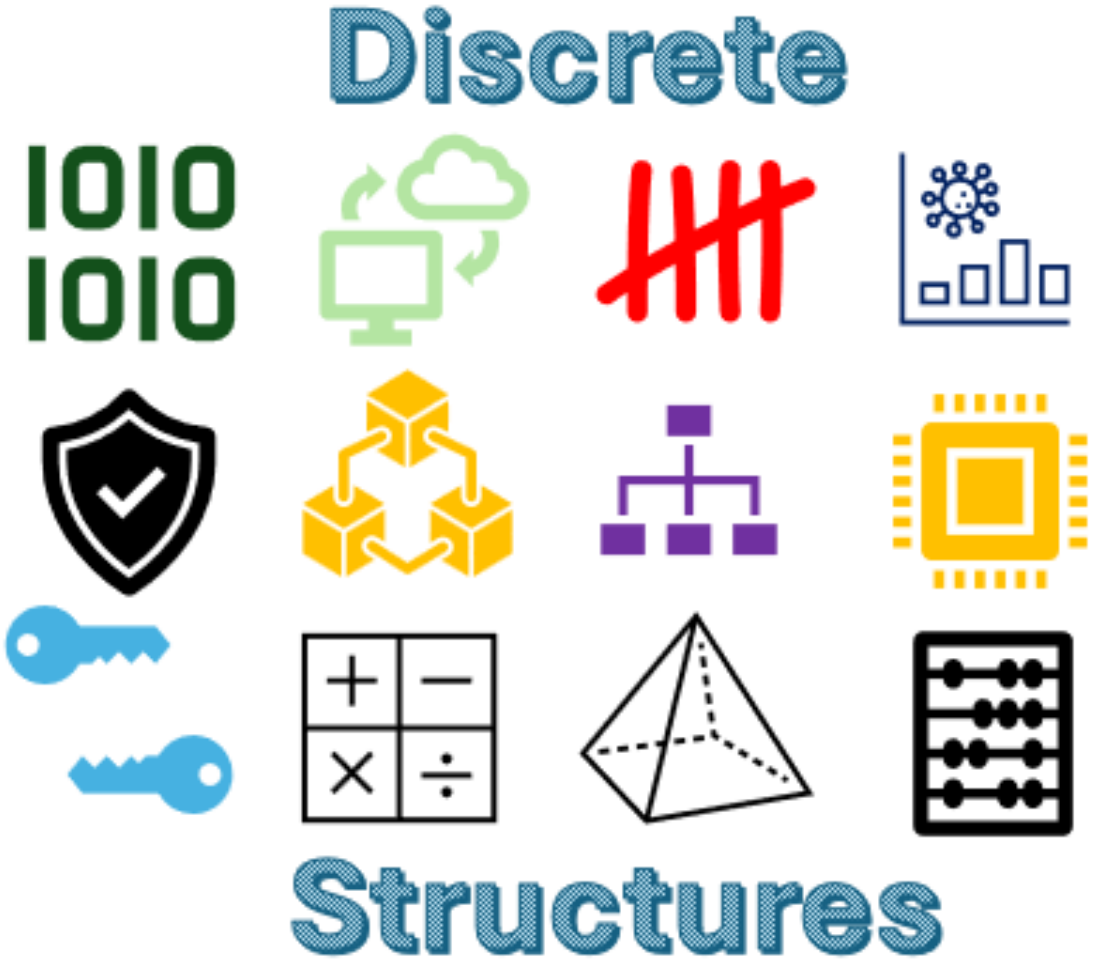
# COMP 2310: Discrete Structures
# Fall 2026

Chapter 2: Logic



# What is logic

- Methods of reasoning (why?, clarification, justification):
  - Rules and techniques to determine whether a given argument is valid
- Application
  - **Mathematics:** to prove theorems
  - **Computer Science:** to verify the correctness of programs
  - **Science:** to draw conclusions from experiments
  - **Social science and daily life:** to problem-solving

# Historical background: Augustus De Morgan

- **Augustus De Morgan** (1806-1871), British mathematician, logician.
- Extended and corrected logic ideas developed by early Greeks.
- First to use the term **“induction”** in formal proofs.
- De Morgan’s work put inductive reasoning on a **rigorous foundation**.



Augustus De Morgan

# Historical background: George Boole

- **George Boole** (1815-1864), British mathematician.
- Published influential works on logic:
  - **The Mathematical Analysis of Logic** (1847)
  - An Investigation of the Laws of Thought
- Objective: To investigate the **laws of reasoning** and express them in a **symbolic language**.
- Established the use of **algebraic symbols** in logic.
  - $P \rightarrow Q$ (if P is true then Q is true)
- Boole's work laid the foundation for **mathematical logic** and its applications in various fields.



George Boole

# Proposition and logical operations

- Proposition or Statement: a declarative statement (not vice-versa) that can be either **True** or **False False,** but not both

**Example 1** Which of the following are statements?

**(a)** The earth is round.
**(b)** $2 + 3 = 5$
**(c)** Do you speak English?
**(d)** $3 - x = 5$
**(e)** Take two aspirins.
**(f)** The temperature on the surface of the planet Venus is 800°F.
**(g)** The sun will come out tomorrow.

- Statements that are proposition (what are the statements that can be clearly evaluated?):
- a, b, f, g

# Logical connectives and compound statements

- In Mathematics:
  - x, y, z are often referred to as variable that can be replaced with values
  - +,×,−, and ÷ represent operation
- In Logic:
  - p, q, r  are often referred to as propositional variables that can be replaced by statements
    - If p: the sun is shining today, q: it is cold
- Logical connectives:
  - p and q can be connected with **logical connectives** (the sun is shining today **and** it is cold)
  - Used to form compound statements

# Logical connectives

- Negation (not)
- Conjunction (and)
- Disjunction (or)
- Implication (if…then)
- Biconditional (if and only if) [p: it is raining, q: the ground is wet. It is raining **if and only if** the ground is wet]

# Truth table of logical connectives

**TABLE 2.1**

| $p$ | $\sim p$ |
|---|---|
| T | F |
| F | T |

Negation

**TABLE 2.2**

| $p$ | $q$ | $p \wedge q$ |
|---|---|---|
| T | T | T |
| T | F | F |
| F | T | F |
| F | F | F |

Conjunction (AND)

**TABLE 2.3**

| $p$ | $q$ | $p \vee q$ |
|---|---|---|
| T | T | T |
| T | F | T |
| F | T | T |
| F | F | F |

Disjunction (OR)

# Logical connectives

- Make a truth table for the statement $(p \wedge q) \vee (\sim p)$
  - How many possible unique combinations = $2^n$ ,where n is the number of variables

| $p$ | $q$ | $p \wedge q$ | $\vee$ | $\sim p$ |
|---|---|---|---|---|
| T | T | T | T | F |
| T | F | F | F | F |
| F | T | F | T | T |
| F | F | F | T | T |

# Quantifiers

- Predicate: A statement P(x) that depends on a variable x
  - A = {*x* | P*(x)*}
    - *A* = {*x* | *x* is an integer less than 8}.
  - P(x) is called **predicate** or **propositional function**
  - Each choice of x produces a proposition P(x) that is either true or false.
  - Application in programming:
    - If P(x): execute specific step(s)
    - While Q(x): do specific step(s)

# Universal quantification

- Statement: For all values of x, P(x) is true
- Notation: ∀xP(x)
  - (a) P(x): -(-x) = x
  - The universal quantification of P(x), ∀x P(x)
- Let Q(x): x + 1 < 4. Then ∀x Q(x) is a false statement, because Q(5) is not true.
- Universal quantification can also be stated in English as “for every *x*,” “every *x*,” or “for any *x*.”

# Existential quantification

- The statement “There exists a value of x for which P(x) is true.
- ∃xP(x) (Existential Quantifier)
  - Let Q*(x)*: *x* +1 < 4. The existential quantification of Q*(x)*, ∃*x* Q*(x)*, is a true statement, because Q(2) is a true statement.
  - The statement ∃*y* *y* + 2 = *y* is false.

# Negation in quantifiers

- Negative a universal statement
  - $\sim\forall x P(x)$ is equivalent to $\exists x \sim P(x)$
- Negative an existential statement
  - $\sim\exists x P(x)$ is equivalent to $\forall x \sim P(x)$
- Examples:

**(a)** Let $p$: For all positive integers $n$, $n^2 + 41n + 41$ is a prime number. Then $\sim p$ is There is at least one positive integer $n$ for which $n^2 + 41n + 41$ is not prime.

**(b)** Let $q$: There is some integer $k$ for which $12 = 3k$. Then $\sim q$: For all integers $k$, $12 \neq 3k$. ◆

# Exercise

*In Exercises 8 and 9, find the truth value of each proposition if $p$ and $r$ are true and $q$ is false.*

**8.** **(a)** $\sim p \wedge \sim q$ **(b)** $(\sim p \vee q) \wedge r$

**(c)** $p \vee q \vee r$ **(d)** $\sim(p \vee q) \wedge r$

**9.** **(a)** $\sim p \wedge (q \vee r)$ **(b)** $p \wedge (\sim(q \vee \sim r))$

**(c)** $(r \wedge \sim q) \vee (p \vee r)$ **(d)** $(q \wedge r) \wedge (p \vee \sim r)$

# Exercise

*For Exercises 31 through 33, define $p \downarrow q$ to be a true statement if neither $p$ nor $q$ is true.*

| $p$ | $q$ | $p \downarrow q$ |
|---|---|---|
| T | T | F |
| T | F | F |
| F | T | F |
| F | F | T |

**31.** Make a truth table for $(p \downarrow q) \downarrow r$.

**32.** Make a truth table for $(p \downarrow q) \wedge (p \downarrow r)$.

**33.** Make a truth table for $(p \downarrow q) \downarrow (p \downarrow r)$.

**Note:** $\uparrow$ represents NAND (Not AND) operation
$\downarrow$ represents NOR (Not OR) operation